João Jerónimo

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Summary

Data Scientist and PhD Economist with 5+ years turning messy multi-source data into models that inform decisions. Built production-style pipelines (Python, SQL, DuckDB) ingesting and reconciling data from APIs, scrapers and administrative records, and modelled 50M+ row panels at scale. Experience across the full modelling stack with a causal-inference background that makes the difference between correlation and effect explicit. Regular experience communicating technical results to non-specialist audiences, in consulting deliverables, conference presentations and teaching.

Work experience

AI Evaluation Expert *Remote*
micro1 *Feb 2026 – Jul 2026*

- Designed domain-specific evaluation frameworks to assess LLM performance on economics and data science tasks, including rubric development for causal inference workflows, econometric method selection, and code quality for OpenAI.
- Designed synthetic datasets with controlled statistical properties (endogeneity, heteroskedasticity, panel structure) to stress-test estimator behaviour and validate experiment design under realistic, adversarial conditions.
- Authored expert-level task specifications simulating senior economist instruction, used to evaluate model capability across regression, forecasting, and causal identification tasks.

Data Scientist (Doctoral Researcher) *Edinburgh, UK*
Heriot-Watt University *Sep 2021 – Aug 2026*

- Modeled price (wage) elasticities of supply on a 50M+ row two-sided panel, showing they move from 2.7 to 0.6 across the cycle; diagnosed a lagging, noisy feedback signal producing sign-flipped estimates and replaced it with a contemporaneous flow signal to restore stable, interpretable results.
- Cut macro forecast error by ≈60% (RMSE) by benchmarking an LSTM pipeline against the incumbent ARIMA model on a UK government forecasting dataset.
- Built a queryable graph of worker flows across 357K firms — a two-sided matching network — enabling causal inference on sparse subsets where standard estimators break down.
- Built a counterfactual simulation engine in Python to forecast agent responses to price and incentive changes, replacing a static policy model with a behavioural, response-based one.
- Communicated complex modelling results to non-specialist audiences, including conference presentations and undergraduate instruction.

Data Research Analyst *Lisbon, Portugal*
University of Lisbon *Jan 2019 – Jun 2021*

- Pre-processing and streamlining data across multiple formats (.csv, .json and HTML)
- Automation of data extraction processes through web scraping using Python, resulting in data preparation time reduction from 5-6 days to under 2 hours.
- Writing VBA scripts for simultaneous operations in large matrices, significantly speeding up data operations required for pre-processing.

Skills

Pricing & marketplace modeling: elasticity & supply/demand response, structural/dynamic optimization, incentive & behavioural modeling, two-sided matching.
Causal inference & experimentation: A/B testing, DiD (incl. staggered adoption), IV/2SLS (weak-instrument diagnostics), Synthetic Control, RDD, Double/Debiased ML (DML).

Modeling: forecasting & time series (ARIMA, LSTM), Bayesian econometrics, high-dimensional/regularised regression (LASSO).

Engineering: Python (Advanced), SQL, R (Advanced), DuckDB, Git/GitHub; production-style data pipelines, reproducible workflows.

Independent projects

Iberian Electricity Market Data Pipeline *github.com/mibel*

- Built a production-style pipeline from scratch to ingest, clean, and model hourly market observations (OMIE, ENTSO-E, Open-Meteo) into a DuckDB warehouse and an analysis-ready panel – to evaluate the Iberian gas-price cap, a real market-design / price-intervention question.
- Tools Used: Python (pandas, requests, entsoe-py), DuckDB, SQL, Open-Meteo API, Git/GitHub, Jupyter

Macroeconomic Forecasting Engine *github.io.com/sfc*

- Built LSTM pipeline, benchmarked against ARIMA to enhance forecasting accuracy of a UK government forecasting model by close to 60% (in terms of RMSE).
- Tools Used: Python (scikit-learn, TensorFlow)

Medium-term Inflation Predictor *github.com/ml*

- Wrote and calibrated ARIMA, VAR and Bayesian UC-SV models to predict quarterly inflation. Pseudo out-of-sample model evaluation, robust standard errors and comparative forecast table.
- Tools Used: Python (numpy, SciPy, pandas)

Education

Heriot-Watt University *Oct 2021 – Jan 2026*
PhD in Economics

- Research areas: macroeconomics, labour economics, applied econometrics, data science. Award: UK Research and Innovation (UKRI) Scholarship (≈£60,000).

University of Minho *Sept 2017 – Jan 2020*
MSc in Economics (summa cum laude)

- Honors: National Award for Academic Merit (Top ranked students in Portugal, 2019); "Best Student" scholarships for highest GPA in 2018.

University of Minho *Sept 2014 – Jun 2017*
BA in International Relations (magna cum laude)

- Honors: "Best Student" scholarships for highest GPA in 2016.

Journal publications

Interactions between financial constraints and economic growth Jul 2023

J. Jerónimo, A. Azevedo, P.C. Neves, M. Thompson

The North American Journal of Economics and Finance, vol. 67

10.1016/j.najef.2023.101943

Working papers

Bias-Corrected Industry Premiums and Wage Floor Effectiveness: Evidence from a Rigid Labour Market 2026

J. Jerónimo, JMP

Working paper